HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed C++ components for real-time data processing systems, integrating with **Linux** environments and adhering to strict coding standards.
- Authored Python scripts to automate system diagnostics and data extraction for embedded system analysis, reducing manual effort by approximately **40%**.
- Contributed to the design and implementation of hardware-software interfaces for embedded devices, focusing on efficient data flow and low-level system interaction.
- Worked with experienced engineers on **embedded systems** and **edge computing** projects, gaining exposure to real-world hardware constraints and performance optimization.
- Participated in code reviews for C++ and Python modules, ensuring adherence to best practices and improving overall code quality and maintainability within the team.
- Gained exposure to **Machine Learning** concepts through team discussions and initial project explorations, focusing on how these models interact with embedded hardware.
- Containerized development environments using **Docker** for consistent build and deployment across multiple developer machines and testing setups.

SKILLS

Programming Languages – C++, Python

Operating Systems – Linux

Embedded Systems – Embedded Systems, Edge Computing, Hardware/Software Interaction

Machine Learning – Machine Learning, Deep Learning, PyTorch, TensorFlow

Software Engineering – Agile Methodologies, Code Reviews, Data Structures, Algorithms, Debugging

PROJECTS

Real-time Data Ingestion for Embedded Devices

- Developed a **C++** application to capture and process sensor data streams from **embedded systems** in real-time.
- Implemented efficient data parsing and serialization logic, enabling data transfer to a central processing unit with minimal latency.
- Integrated the system with **Linux**-based firmware, ensuring compatibility and stability across different hardware configurations.

Automated Diagnostic Tool for IoT Fleet

- Created a **Python**-based tool to automatically collect diagnostic logs and system status from a fleet of IoT devices.
- Designed the tool to identify common failure patterns and generate summary reports, improving troubleshooting efficiency by **~35%**.
- Focused on robust error handling and secure data aggregation for eventual analysis on the edge.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)